Tennis Project Concise Technical Summary

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1 Data, Prediction Targets and Feature Engineering

The project aims to predict a professional men's tennis match fairly accurately, before it starts, through a two-stage framework. Machine Learning models first estimate a player's probability to win a point on a serve. These probabilities are then propagated through an analytical representation of tennis scoring to obtain game, set and match winning probabilities.

The data used for this project was collected from Jeff Sackmann's public `tennis_atp` repository. Three groups of annual files were downloaded for 1991–2026: main-tour matches, Futures matches, and qualifying/Challenger matches. Their union contained 855,194 match records. The primary modeling sample is smaller because detailed serve statistics are not available for every record.

The original winner/loser representation was converted into a Player A/Player B representation using a reproducible random assignment. This prevents a model from learning the artificial rule that the first player is always the winner.

For a particular match, the two regression targets for the ML models are the realized proportions of service points won:

$$p_A = \frac{W_{1,A} + W_{2,A}}{N_A}, \qquad p_B = \frac{W_{1,B} + W_{2,B}}{N_B}, \tag{1}$$

$$\text{where} \quad \begin{cases} W_{1,i} & \text{first-serve points won by player } i, \\ W_{2,i} & \text{second-serve points won by player } i, \\ N_i & \text{total service points played by player } i. \end{cases}$$

Matches were retained only when these quantities were available with positive denominators for both players. This produced 259,423 usable matches. However, the other matches weren't just discarded- information from those matches was still used to create features, which were fed to the ML models to predict point-winning probabilities.

The feature set contains approximately 200 pre-match variables, expanding to 247 columns after categorical encoding. These include ATP rankings and ranking points; overall, surface-specific, blended, momentum, and volatility variants of Elo; recent form; career and surface records; head-to-head performance; workload and fatigue; historical serve and return statistics; player age and height; tournament experience; seeding; playing surface; tournament level; round; and match format.

Historical features were generated sequentially. For each match, the existing player histories were read first, the pre-match feature row was recorded, and only then were the histories updated using the result and statistics of that match. Thus, current-match serve statistics determine the regression labels but are not used as predictors for the same match.

2 Analytical Tennis Scoring Model

Let `p` be the estimated probability that Player A wins a point on serve and `q` the corresponding probability for Player B. Assuming independent points and a constant serve-point probability during the match, the probability that a player with serve probability `x` wins a service game is:

$$G(x) = x^4 + 4(1-x)x^4 + 10(1-x)^2x^4 + 20(1-x)^3x^3\left[\frac{x^2}{1-2x(1-x)}\right]. \tag{2}$$

The first three terms describe winning before deuce. The final term is the probability of reaching deuce multiplied by the probability of eventually winning the two-point deuce process.

For Player A,

$$t = G(p) \tag{3}$$

is the probability of holding serve, while

$$r = 1 - G(q) \tag{4}$$

is the probability of breaking Player B's serve.

The set calculation enumerates the valid winning scorelines:

$$P_{\text{set}} = P_{6\text{–}0} + P_{6\text{–}1} + P_{6\text{–}2} + P_{6\text{–}3} + P_{6\text{–}4} + P_{7\text{–}5} + P_{7\text{–}6}. \tag{5}$$

Each scoreline probability is constructed from the appropriate combinations of service holds and breaks. The 7–6 term separately accounts for reaching 6–6 and winning the tiebreak. The complete expressions are implemented in the analytical scoring code, and have been listed in the `probability_calculations` PDF.

If $s = P_{\text{set}}$, match-winning probabilities are

$$P_{\text{Bo3}} = s^2 + 2s^2(1 - s) = 3s^2 - 2s^3 \tag{6}$$

for a best-of-three match, and

$$P_{\text{Bo5}} = s^3 + 3s^3(1 - s) + 6s^3(1 - s)^2 \tag{7}$$

for a best-of-five match.

The model's main simplifying assumptions are that serve-point probabilities remain constant during a match; successive points, games, and sets are conditionally independent; and unexpected in-match effects such as injury, weather, pressure, fatigue, and tactical adaptation are not modelled dynamically. These assumptions provide analytical interpretability but limit the model's ability to represent changing match conditions.

3 Training, Validation, Testing, and Leakage Control

After target availability filtering, the data were retained in chronological order and divided into:

- **Training:** 181,596 matches (70%), ending approximately July 2022.
- **Validation:** 25,942 matches (10%), approximately July 2022–July 2024.
- **Test:** 51,885 matches (20%), approximately July 2024–June 2026.

The training data were used for model fitting, while the validation period was used for model and ensemble selection. Time-aware expanding-window cross-validation was used during hyperparameter search so that each validation fold occurred after its corresponding training fold.

Seven serve-probability regressors were investigated: Ridge, Elastic Net, Random Forest, Extra Trees, XGBoost, LightGBM, and CatBoost. Ridge regression provided the principal simple linear baseline. Equal-weight, validation-error-weighted, and stacked ensembles were then examined. The selected model was an equal-weight ensemble of Ridge, LightGBM, XGBoost, and CatBoost, which achieved the lowest validation MAE among the evaluated ensembles.

At the match-probability level, a constant 0.5 prediction provides the simplest reference, corresponding to log loss (0.693), Brier score (0.25), and ROC-AUC (0.5). As part of the final reproducibility review, I also plan to report higher-ranked-player and pre-match-Elo baselines so that the hybrid framework can be compared against practical tennis-specific alternatives.

4 Preliminary Out-of-Sample Results and Interpretation

On the chronological test period, the serve-probability ensemble obtained an average MAE of approximately 0.06715. Passing its estimates through the analytical scoring model produced the following preliminary results:

These results show that the framework contains predictive information beyond a constant 50–50 forecast and has useful discriminatory ability. They do not, by themselves, establish superiority to every tennis-specific baseline or imply that individual match outcomes are predictable with certainty.

Metric	Result
Accuracy	0.6958
Log loss	0.6156
Brier score	0.2032
ROC-AUC	0.7666

A ten-bin reliability diagram indicated reasonably close agreement between predicted and observed frequencies through much of the central probability range, with larger deviations at the extremes. Consequently, the current evidence supports the absence of severe central-range miscalibration, but not a claim of perfect calibration.

The clearest error pattern concerns closely matched players. Test accuracy was approximately 58.4% when the players' overall Elo ratings differed by fewer than 25 points and 60.1% for differences of 25–50 points. It increased to 74.2% for differences of 100–200 points and approximately 86.5% beyond 200 points. This indicates that most remaining errors are concentrated in inherently competitive matches where the pre-match information does not strongly distinguish the players.

Further work will aim to study close matches separately. Other than experimenting with simulation noise, I also plan to estimate uncertainty around p and q and propagate that uncertainty through the scoring model. This would allow a close match to be represented by a distribution of feasible match probabilities rather than a single over-precise estimate. I also plan to examine the 40–60% match-probability region using richer player–opponent interactions, confidence-conditioned calibration, and rolling retraining as new match information becomes available.

A longer-term extension is to relax the constant-(p, q) assumption. A state-dependent simulator could allow serve effectiveness to vary with set state, accumulated fatigue, recent performance within the match, or pressure situations. A corrected high-trial Monte Carlo implementation could be used to support these more dynamic models when closed-form calculation is no longer practical.

The present figures are preliminary results from an ongoing project. The modelling pipeline has undergone chronological, diagnostic, interpretability, calibration, and subgroup analyses during development, but the final reported values remain subject to minor changes following one last reproducibility and consistency review.